

# SQL



## Beginner & Intermediate Challenge

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1:23



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# Dataset and Tools

## Overview

An open and enhanced dataset of musical elements (music, albums, and artists) suitable for music data mining. The attractive features of MusicOSet include the enrichment of existing metadata to which it is linked and the popularity classification of the musical elements present in the dataset.

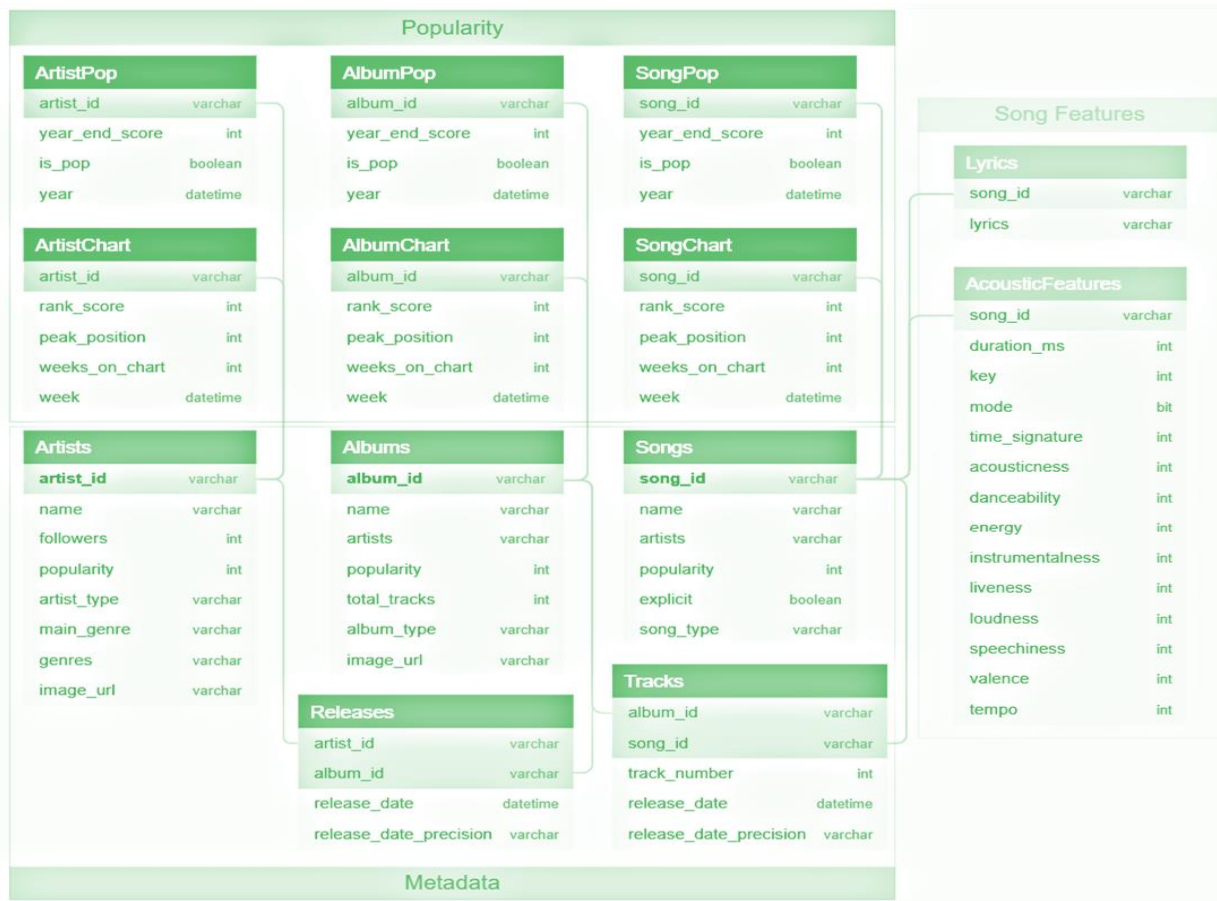


An Enhanced Music Dataset  
for Music Data Mining

BigQuery is a serverless, cost-effective, and multi-cloud data warehouse designed to help companies turn big data into valuable business insights.



# Table Scheme



# Beginner SQL

# Question 1

[>> SQL Query Syntax Link for Question 1](#)

Find all the information about each artist.

```
SELECT  
*  
FROM  
`exercise-347601.music_dataset.artists`
```

# Question 1

Find all the information about each artist.

## Query Result

Table Schema

| Field name  | Type    | Mode     |
|-------------|---------|----------|
| artist_id   | STRING  | NULLABLE |
| name        | STRING  | NULLABLE |
| followers   | INTEGER | NULLABLE |
| popularity  | INTEGER | NULLABLE |
| artist_type | STRING  | NULLABLE |
| main_genre  | STRING  | NULLABLE |
| genres      | STRING  | NULLABLE |
| image_url   | STRING  | NULLABLE |

Table Preview

| Row | artist_id              | name                | followers | popularity | artist_type | main_genre  | genres                                        |
|-----|------------------------|---------------------|-----------|------------|-------------|-------------|-----------------------------------------------|
| 1   | 66CXWjxzNUsdJxJ2JdwvnR | Ariana Grande       | 34554242  | 96         | singer      | dance pop   | ['dance pop', 'pop', 'post-teen pop']         |
| 2   | 26VFTg2z8YR0cCuwLzESi2 | Halsey              | 7368242   | 90         | singer      | dance pop   | ['dance pop', 'electropop', 'etherpop', ']    |
| 3   | 0Y5tJX1MQIPqiwIOH1tJY  | Travis Scott        | 6313709   | 94         | rapper      | pop         | ['pop', 'pop rap', 'rap']                     |
| 4   | 246dkjvS1zLTtiykXe5h60 | Post Malone         | 16737002  | 96         | rapper      | dfw rap     | ['dfw rap', 'pop', 'rap']                     |
| 5   | 1zNqQNideOUZHb8zbZRFMX | Swae Lee            | 483032    | 89         | singer      | trap music  | ['trap music']                                |
| 6   | 64KEffDW9EtZ1y2vBYgq8T | Marshmello          | 15566666  | 91         | DJ          | brostep     | ['brostep', 'progressive electro house        |
| 7   | 7EQ0qTo7fWT7DPxmtSYEc  | Bastille            | 3776115   | 85         | band        | metropolis  | ['metropolis', 'modern rock', 'pop', ']       |
| 8   | 20JZFwl6HVI6yg8a4H3ZqK | Panic! At The Disco | 7749228   | 88         | band        | baroque pop | ['baroque pop', 'emo', 'modern rock', ']      |
| 9   | 4iHNK0tOyZPYnBU7nGAgpQ | Mariah Carey        | 4682308   | 81         | singer      | dance pop   | ['dance pop', 'pop', 'r&b', 'urban conte      |
| 10  | 4DdkRBBYG6Yk9Ka8tdJ9BW | Offset              | 1202706   | 89         | singer      | atl hip hop | ['atl hip hop', 'pop', 'pop rap', 'rap', 'sou |
| 11  | 46SHBwWsqBkxI7EeeBEQG7 | Kodak Black         | 5001866   | 86         | singer      | florida rap | ['florida rap', 'hip hop', 'miami hip hop     |
| 12  | 5f7VJfbbwm532GiveGC0ZK | Lil Baby            | 2571569   | 90         | rapper      | atl hip hop | ['atl hip hop', 'atl trap', 'rap', 'trap musi |
| 13  | 2hlmm7s2ICUX0LVIhVFIZQ | Gunna               | 1298099   | 88         | singer      | atl hip hop | ['atl hip hop', 'rap', 'southern hip hop',    |

## Question 2

[>> SQL Query Syntax Link for Question 2](#)

Find artist name and followers with a popularity of 90.

```
SELECT
  name,
  followers
FROM
  `exercise-347601.music_dataset.artists`
WHERE
  popularity = 90
```

## Question 2

Find artist name and followers with a popularity of 90.

## Query Result

Table Schema

| Field name | Type    | Mode     |
|------------|---------|----------|
| name       | STRING  | NULLABLE |
| followers  | INTEGER | NULLABLE |

Table Preview

| Row | name                   | followers |
|-----|------------------------|-----------|
| 1   | Halsey                 | 7368242   |
| 2   | Lil Baby               | 2571569   |
| 3   | Bruno Mars             | 22140623  |
| 4   | Future                 | 7985083   |
| 5   | A Boogie Wit da Hoodie | 3276918   |
| 6   | Chance the Rapper      | 4307572   |
| 7   | Ty Dolla Sign          | 2875582   |
| 8   | Kygo                   | 5354354   |
| 9   | The Weeknd             | 16909290  |
| 10  | Kendrick Lamar         | 13025240  |



# Question 3

[>> SQL Query Syntax Link for Question 3](#)

Find pop artists with between 1000 and 10,000 followers.

```
SELECT
  name,
  followers,
  popularity
FROM
  `exercise-347601.music_dataset.artists`
WHERE
  followers NOT BETWEEN 1000
  AND 10000
  AND main_genre = 'pop'
```

## Question 3

Find pop artists with between 1000 and 10,000 followers.

### Query Result

Table Schema

| Field name | Type    | Mode     |
|------------|---------|----------|
| name       | STRING  | NULLABLE |
| followers  | INTEGER | NULLABLE |
| popularity | INTEGER | NULLABLE |

Table Preview

| Row | name              | followers | popularity |
|-----|-------------------|-----------|------------|
| 1   | Christopher Wilde | 89482     | 57         |
| 2   | R5                | 540136    | 59         |
| 3   | The Carters       | 610316    | 66         |
| 4   | Riverdale Cast    | 373552    | 67         |
| 5   | Grace VanderWaal  | 1099173   | 68         |
| 6   | dodie             | 531224    | 69         |
| 7   | Rachel Platten    | 768258    | 70         |

# Question 4

[>> SQL Query Syntax Link for Question 4](#)

List the last five non-pop artists and their followers, arranged alphabetically, whose names begin with the letter 'A'.

```
SELECT
  name,
  followers,
  artist_type
FROM `exercise-347601.music_dataset.artists`
WHERE name LIKE "A%" AND main_genre != 'pop'
ORDER BY followers DESC
LIMIT 5
```

# Question 4

List the last five non-pop artists and their followers, arranged alphabetically, whose names begin with the letter 'A'.

## Query Result

| Table Schema |         |          |
|--------------|---------|----------|
| Field name   | Type    | Mode     |
| name         | STRING  | NULLABLE |
| followers    | INTEGER | NULLABLE |
| artist_type  | STRING  | NULLABLE |

| Table Preview |               |           |             |
|---------------|---------------|-----------|-------------|
| Row           | name          | followers | artist_type |
| 1             | Ariana Grande | 34554242  | singer      |
| 2             | Adele         | 17202908  | singer      |
| 3             | Avicii        | 14748479  | DJ          |
| 4             | Alan Walker   | 12941844  | DJ          |
| 5             | AC/DC         | 12366680  | band        |

## Question 5

>> [SQL Query Syntax Link for Question 5](#)

Listing the top 50 popular pop, hip hop, and k-pop artists (name, followers, main\_genre, popularity\_rank, category).

```
SELECT
  name,
  followers,
  main_genre,
  popularity AS popularity_rank,
  CASE
    WHEN popularity <= 50 THEN 'Top Popular 50 Artist'
    WHEN popularity > 50 AND popularity <= 100 THEN 'Second Place Popular Artist'
    ELSE 'Others' END AS category
FROM
  `exercise-347601.music_dataset.artists`
WHERE
  main_genre IN ('pop', 'hip hop', 'k-pop')
ORDER BY
  followers
```

# Question 5

Listing the top 50 popular pop, hip hop, and k-pop artists (name, followers, main\_genre, popularity\_rank, category).

## Query Result

Table Schema

| Field name      | Type    | Mode     |
|-----------------|---------|----------|
| name            | STRING  | NULLABLE |
| followers       | INTEGER | NULLABLE |
| main_genre      | STRING  | NULLABLE |
| popularity_rank | INTEGER | NULLABLE |
| category        | STRING  | NULLABLE |

Table Preview

| Row | name              | followers | main_genre | popularity_rank | category                    |
|-----|-------------------|-----------|------------|-----------------|-----------------------------|
| 1   | Salaam Remi       | 11596     | hip hop    | 67              | Second Place Popular Artist |
| 2   | Audio Two         | 17665     | hip hop    | 31              | Top Popular 50 Artist       |
| 3   | Ben               | 19622     | k-pop      | 53              | Second Place Popular Artist |
| 4   | Skee-Lo           | 32248     | hip hop    | 57              | Second Place Popular Artist |
| 5   | K-OS              | 36026     | hip hop    | 46              | Top Popular 50 Artist       |
| 6   | Grey              | 40557     | pop        | 77              | Second Place Popular Artist |
| 7   | Wreckx-N-Effect   | 53211     | hip hop    | 45              | Top Popular 50 Artist       |
| 8   | Guru's Jazzmatazz | 55450     | hip hop    | 47              | Top Popular 50 Artist       |
| 9   | Young MC          | 78288     | hip hop    | 54              | Second Place Popular Artist |

# Intermediate SQL

# Question 6

[>> SQL Query Syntax Link for Question 6](#)

Create a query to calculate the overall number of album releases every month.

```
SELECT
  EXTRACT (YEAR FROM release_date) AS year,
  EXTRACT (MONTH FROM release_date) AS month,
  COUNT(album_id) AS album_count
FROM
  `exercise-347601.music_dataset.releases`
WHERE
  release_date BETWEEN '2010-01-01' AND '2020-12-31'
GROUP BY year, month
ORDER BY year DESC, month
```



# Question 6

Create a query to calculate the overall number of album releases every month.

## Query Result

Table Schema

| Field name  | Type    | Mode     |
|-------------|---------|----------|
| year        | INTEGER | NULLABLE |
| month       | INTEGER | NULLABLE |
| album_count | INTEGER | NULLABLE |

Table Preview

| Row | year | month | album_count |
|-----|------|-------|-------------|
| 1   | 2020 | 1     | 83          |
| 2   | 2020 | 3     | 64          |
| 3   | 2020 | 6     | 79          |
| 4   | 2020 | 7     | 86          |
| 5   | 2020 | 11    | 67          |
| 6   | 2019 | 1     | 24          |

# Question 7

[>> SQL Query Syntax Link for Question 7](#)

Create a query to find the top ten non-various artists with the most albums and the average album track number.

```
SELECT
  artists.name AS artist_name,
  COUNT(albums.name) AS albums_numbers,
  ROUND(AVG(albums.total_tracks),2) AS average_tracks
FROM
  `exercise-347601.music_dataset.artists` AS artists
LEFT JOIN
  `exercise-347601.music_dataset.albums` AS albums
ON
  albums.artists_id = artists.artist_id
WHERE artists.name != 'Various Artists' AND albums.name IS NOT NULL
GROUP BY artists.name
ORDER BY albums_numbers DESC
LIMIT 10
```

# Question 7

Create a query to find the top ten non-various artists with the most albums and the average album track number.

## Query Result

Table Schema

| Field name     | Type    | Mode     |
|----------------|---------|----------|
| artist_name    | STRING  | NULLABLE |
| albums_numbers | INTEGER | NULLABLE |
| average_tracks | FLOAT   | NULLABLE |

Table Preview

| Row | artist_name              | albums_numbers | average_tracks |
|-----|--------------------------|----------------|----------------|
| 1   | The Karaoke Channel      | 116            | 2.54           |
| 2   | Mega Tracks Karaoke Band | 76             | 1.93           |
| 3   | Bob Dylan                | 55             | 16.07          |
| 4   | Barbra Streisand         | 52             | 14.88          |
| 5   | Willie Nelson            | 51             | 14.24          |
| 6   | Elvis Presley            | 49             | 25.71          |
| 7   | The Rolling Stones       | 42             | 15.74          |
| 8   | Johnny Mathis            | 42             | 12.9           |
| 9   | The Temptations          | 41             | 10.95          |
| 10  | Neil Young               | 40             | 11.28          |

# Question 8

>> [SQL Query Syntax Link for Question 8](#)

Create a query to retrieve a list of songs whose album names begin with the letter 'A' and were released between 2010 and 2020 (song\_name, artist\_name, album\_name, release\_year).

```
SELECT
songs.song_name AS song_name,
artists.name AS artist_name,
albums.name AS album_name,
EXTRACT (YEAR FROM releases.release_date) AS release_year
FROM `exercise-347601.music_dataset.songs` AS songs
LEFT JOIN `exercise-347601.music_dataset.artists` AS artists
ON songs.main_artist_id = artists.artist_id
LEFT JOIN `exercise-347601.music_dataset.tracks` AS tracks
ON songs.song_id = tracks.song_id
LEFT JOIN `exercise-347601.music_dataset.albums` AS albums
ON tracks.album_id = albums.album_id
LEFT JOIN `exercise-347601.music_dataset.releases` AS releases
ON albums.album_id = releases.album_id
WHERE releases.release_date BETWEEN '2010-01-01' AND '2020-12-31' AND albums.name LIKE 'A%'
ORDER BY songs.popularity DESC
```

## Question 8

Create a query to retrieve a list of songs whose album names begin with the letter 'A' and were released between 2010 and 2020 (song\_name, artist\_name, album\_name, release\_year).

### Query Result

Table Schema

| Field name   | Type    | Mode     |
|--------------|---------|----------|
| song_name    | STRING  | NULLABLE |
| artist_name  | STRING  | NULLABLE |
| album_name   | STRING  | NULLABLE |
| release_year | INTEGER | NULLABLE |

Table Preview

| Row | song_name          | artist_name      | album_name                      | release_year |
|-----|--------------------|------------------|---------------------------------|--------------|
| 1   | SICKO MODE         | Travis Scott     | ASTROWORLD                      | 2018         |
| 2   | Young Dumb & Broke | Khalid           | American Teen                   | 2017         |
| 3   | Location           | Khalid           | American Teen                   | 2017         |
| 4   | Do I Wanna Know?   | Arctic Monkeys   | AM                              | 2013         |
| 5   | Redbone            | Childish Gambino | Awaken, My Love!                | 2016         |
| 6   | YOSEMITE           | Travis Scott     | ASTROWORLD                      | 2018         |
| 7   | BUTTERFLY EFFECT   | Travis Scott     | ASTROWORLD                      | 2018         |
| 8   | STARGAZING         | Travis Scott     | ASTROWORLD                      | 2018         |
| 9   | Centuries          | Fall Out Boy     | American Beauty/American Psycho | 2015         |

# Question 9

[>> SQL Query Syntax Link for Question 9](#)

Create a query to find songs that stayed more than two weeks on the charts and reached more than two peak positions between 2010 and 2020 (song\_name, artist\_name, album\_name, track\_number).

```
SELECT
songs.song_name AS song_name,
artists.name AS artist_name,
albums.name AS album_name,
tracks.track_number AS track_number
FROM `exercise-347601.music_dataset.songs` AS songs
LEFT JOIN `exercise-347601.music_dataset.artists` AS artists
ON songs.main_artist_id = artists.artist_id
LEFT JOIN `exercise-347601.music_dataset.tracks` AS tracks
ON songs.song_id = tracks.song_id
LEFT JOIN `exercise-347601.music_dataset.albums` AS albums
ON tracks.album_id = albums.album_id
WHERE songs.song_id IN (
  SELECT song_id
  FROM `exercise-347601.music_dataset.song_chart`
  WHERE weeks_on_chart > 2 AND week BETWEEN '2010-01-01'
  AND '2020-12-31')
AND tracks.album_id IN (
  SELECT album_id
  FROM `exercise-347601.music_dataset.album_chart`
  WHERE peak_position > 2 AND date BETWEEN '2010-01-01'
  AND '2020-12-31')
ORDER BY songs.song_name
```

## Question 9

Create a query to find songs that stayed more than two weeks on the charts and reached more than two peak positions between 2010 and 2020 (song\_name, artist\_name, album\_name, track\_number).

# Query Result

Table Schema

| Field name   | Type    | Mode     |
|--------------|---------|----------|
| song_name    | STRING  | NULLABLE |
| artist_name  | STRING  | NULLABLE |
| album_name   | STRING  | NULLABLE |
| track_number | INTEGER | NULLABLE |

Table Preview

| Row | song_name                                            | artist_name  | album_name                              | track_number |
|-----|------------------------------------------------------|--------------|-----------------------------------------|--------------|
| 1   | (If You're Wondering If I Want You To) I Want You To | Weezer       | Raditude (International Deluxe Version) | 1            |
| 2   | (Kissed You) Good Night                              | Gloriana     | A Thousand Miles Left Behind            | 4            |
| 3   | ...Ready For It?                                     | Taylor Swift | reputation                              | 1            |
| 4   | 10 Bands                                             | Drake        | If You're Reading This It's Too Late    | 3            |
| 5   | 19 You + Me                                          | Dan + Shay   | Where It All Began                      | 3            |
| 6   | 1901                                                 | Phoenix      | Wolfgang Amadeus Phoenix                | 2            |
| 7   | 1994                                                 | Jason Aldean | Night Train                             | 10           |
| 8   | 1999                                                 | Prince       | 1999                                    | 1            |
| 9   | 2 Phones                                             | Kevin Gates  | Islah (Deluxe)                          | 3            |

# Question 10

[>> SQL Query Syntax Link for Question 10](#)

Create a query to list all explicit songs that have charted for more than two weeks between 2010 and 2020 (song\_name, duration\_second, artist\_name, main\_genre).

```
WITH
explicit_song AS (
  SELECT song_id, song_name, main_artist_id, explicit
  FROM `exercise-347601.music_dataset.songs`
  WHERE explicit = TRUE),
song_weeks_on_chart AS (
  SELECT DISTINCT song_id,
  FROM `exercise-347601.music_dataset.song_chart`
  WHERE weeks_on_chart > 2 AND week BETWEEN '2010-01-01'
  AND '2020-12-31' )

SELECT
  es.song_name,
  ROUND((mf.duration_ms/1000),2) AS duration_second,
  albums.name AS album_name,
  artists.name AS artist_name,
  artists.main_genre
```

```
FROM
  explicit_song AS es
LEFT JOIN `exercise-347601.music_dataset.music_features` AS mf
ON es.song_id = mf.song_id
LEFT JOIN song_weeks_on_chart AS swnc
ON es.song_id = swnc.song_id
LEFT JOIN `exercise-347601.music_dataset.artists` AS artists
ON es.main_artist_id = artists.artist_id
LEFT JOIN `exercise-347601.music_dataset.tracks` AS tracks
ON es.song_id = tracks.song_id
LEFT JOIN `exercise-347601.music_dataset.albums` AS albums
ON tracks.album_id = albums.album_id
WHERE
  albums.name IS NOT NULL AND artists.main_genre IS NOT NULL
```



# Question 10

Create a query to list all explicit songs that have charted for more than two weeks between 2010 and 2020 (song\_name, duration\_second, artist\_name, main\_genre).

## Query Result

Table Schema

| Field name      | Type   | Mode     |
|-----------------|--------|----------|
| song_name       | STRING | NULLABLE |
| duration_second | FLOAT  | NULLABLE |
| album_name      | STRING | NULLABLE |
| artist_name     | STRING | NULLABLE |
| main_genre      | STRING | NULLABLE |

Table Preview

| Row | song_name           | duration_second | album_name              | artist_name   | main_genre  |
|-----|---------------------|-----------------|-------------------------|---------------|-------------|
| 1   | SICKO MODE          | 312.82          | ASTROWORLD              | Travis Scott  | pop         |
| 2   | Mo Bamba            | 183.91          | MUDBOY                  | Sheck Wes     | rap         |
| 3   | breathin            | 198.16          | Sweetener               | Ariana Grande | dance pop   |
| 4   | Better Now          | 231.27          | beerbongs & bentleys    | Post Malone   | dfw rap     |
| 5   | Lucid Dreams        | 239.84          | Goodbye & Good Riddance | Juice WRLD    | chicago rap |
| 6   | Calling My Spirit   | 152.0           | Dying To Live           | Kodak Black   | florida rap |
| 7   | Uproar              | 194.18          | Tha Carter V            | Lil Wayne     | hip hop     |
| 8   | Armed And Dangerous | 170.0           | Goodbye & Good Riddance | Juice WRLD    | chicago rap |
| 9   | Pure Cocaine        | 154.02          | Street Gossip           | Lil Baby      | atl hip hop |



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CREDITS: This presentation was created by **Slidesgo**, including icons by **Flaticon** and infographics & images by **Freepik**

